Ps 05

Erik An

<2026-01-13 Tue>

Contents

1 Problem 1 **DONE** **3**

1.1 Task 3

1.2 Solution 3

1.2.1 Code 3

1.2.2 Solution 3

2 Problem 2 **DONE** **4**

2.1 Task 4

2.2 Notes 4

2.3 Solution 4

2.3.1 Code 4

2.3.2 Solution 6

3 Problem 3 **DONE** **7**

3.1 Task 7

3.2 Solution 7

3.2.1 Code 7

3.2.2 Solution 7

4 Problem 4 **DONE** **8**

4.1 Task 8

4.2 Solution 8

4.2.1 Code 8

4.2.2 Solution 9

5 Problem 5 DONE 10

5.1 Task 10

5.2 Solution 10

5.2.1 (a) Express Area $A(x)$ 10

5.2.2 (b) Determine Maximal Area 10

1 Problem 1 DONE

1.1 Task

Visualize the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = e^{2x} + x$ in Julia. Then prove that f is invertible and determine $(f^{-1})'(1)$.

1.2 Solution

1.2.1 Code

```
using Plots

f(x) = exp(2*x) + x

x = -10:0.1:1
y = f.(x)

plot(x, y,
     title="f(x) = e^(2x) + x",
     xlabel="x",
     ylabel="y",
     linewidth=2)

savefig("./figs/p_01.png")
```

1.2.2 Solution

$$f(x) = e^{2x} + x$$
$$f'(x) = 2e^{2x} + 1$$

Since $\forall x \in \mathbb{R}, f'(x) > 0$ (as $\forall x \in \mathbb{R}, e^x > 0$), $f(x)$ is monotonically increasing (injective).

The function $g : \mathbb{R} \to \mathbb{R}, g(x) = e^{2x}$ is continuous.

The function $k : \mathbb{R} \to \mathbb{R}, k(x) = x$ is also continuous.

Hence the function $f : \mathbb{R} \to \mathbb{R}, f(x) = g(x) + k(x) = e^{2x} + x$ is continuous.

Since $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to\infty} f(x) = \infty$, and using the fact that $f(x)$ is continuous, by the intermediate value theorem $f(x)$ is surjective.

Given $f(x)$ is both injective and surjective, it is also bijective, which implies $f(x)$ is invertible.

$$(f^{-1})'(y) = \frac{1}{f'(x)} = \frac{1}{f'(f^{-1}(y))}$$

We calculate the specific value:

$$\begin{aligned}
1 &= e^{2x} + x \\
x &= 0 \\
(f^{-1})'(1) &= \frac{1}{2e^{2\cdot 0} + 1} \\
&= \frac{1}{2 \cdot 1 + 1} \\
&= \frac{1}{3}
\end{aligned}$$

2 Problem 2 DONE

2.1 Task

Consider the function $f : [1, e] \to \mathbb{R}$, $f(x) = \ln(x)$. Determine ξ associated to the mean value theorem. Illustrate your results in Julia.

2.2 Notes

Plan:

1. Prove $f(x)$ is continuous on $[1, e]$.
2. Prove $f(x)$ is differentiable on $(1, e)$.
3. Apply *Mean Value Theorem.*
4. Solve for ξ.
5. Verify ξ lies in $(1, e)$.

2.3 Solution

2.3.1 Code

```
using Plots
```

```
# define the function and parameters
f(x) = log(x)
a = 1.0
b = exp(1) # e

# define the Solution (xi) you found mathematically
xi = b - 1  #  = e - 1

# calculate slopes and line equations
# slope of the Secant line (Average rate of change)
m_secant = (f(b) - f(a)) / (b - a)

# define the Secant Line function: y = f(a) + m * (x - a)
secant_line(x) = f(a) + m_secant * (x - a)

# define the Tangent Line function at xi: y = f(xi) + f'(xi) * (x - xi)
tangent_line(x) = f(xi) + m_secant * (x - xi)

# generate data for plotting
x_vals = range(a, b, length=100)
y_vals = f.(x_vals)

# plotting
# plot the main function
plot(x_vals, y_vals,
     label="f(x) = ln(x)",
     linewidth=3,
     legend=:bottomright,
     title="Mean Value Theorem Illustration",
     xlabel="x", ylabel="y")

# plot the Secant Line (connecting endpoints)
plot!(x_vals, secant_line.(x_vals),
      label="Secant Line",
      linestyle=:dash,
      linewidth=2)

# plot the Tangent Line (at xi)
plot!(x_vals, tangent_line.(x_vals),
      label="Tangent at ",
      linewidth=2)

# add a marker for the point xi
scatter!([xi], [f(xi)],
         label="Point (, f())",
         markersize=5)

plot!([xi, xi], [0, f(xi)], label="", linestyle=:dot, color=:grey)
```

```
savefig("./figs/p_02.png")
```

2.3.2 Solution

Mean Value Theorem. If $f : [a, b] \to \mathbb{R}$ is continuous and differentiable on (a, b), then there is $\xi \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(\xi)$$

1. Prove $f(x)$ is continuous on $[1, e]$.

   The natural logarithmic function $f(x) = \log(x)$ is defined and continuous $\forall x > 0$. Since $[1, e] \subset (0, \infty)$, $f(x)$ is continuous on $[1, e]$.

2. Prove $f(x)$ is differentiable on $(1, e)$.

   $f'(x) = \dfrac{1}{x}$ is defined $\forall x \neq 0$ and $0 \notin [1, e]$, which implies $f(x)$ is differentiable on $(1, e)$.

3. Apply *Mean Value Theorem.*

$$\frac{f(b) - f(a)}{b - a} = f'(\xi)$$

$$\begin{aligned}
\frac{1}{\xi} &= \frac{\log(e) - \log(1)}{e - 1} && \text{(substituting values)} \\
\frac{1}{\xi} &= \frac{1 - 0}{e - 1} && \text{(simplification)} \\
\frac{1}{\xi} &= \frac{1}{e - 1}
\end{aligned}$$

4. Solve for ξ.

$$\begin{aligned}
\frac{1}{\xi} &= \frac{1}{e - 1} \\
\xi &= e - 1 \\
\xi &\approx 1.7183...
\end{aligned}$$

5. Verify ξ lies in $(1, e)$.

$$1 < 1.7183 \cdots < 2.718 \ldots$$

   Hence, ξ is valid.

3 Problem 3 DONE

3.1 Task

Visualize the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 - 3x^2 + 2$ in Julia and guess its local extrema. Prove that your guess is correct.

3.2 Solution

3.2.1 Code

```
using Plots

f(x) = x^3 - 3x^2 + 2

# Plotting range covering the interesting features
x = -1.5:0.05:3.5
y = f.(x)

p = plot(x, y,
     title = "f(x) = x^3 - 3x^2 + 2",
     xlabel = "x",
     ylabel = "y",
     label = "f(x)",
     linewidth = 2,
     legend = :topright)

# Mark the guessed extrema for visualization
scatter!([0, 2], [f(0), f(2)], label="Extrema guesses", color=:red)

savefig("./figs/p_03.png")
```

3.2.2 Solution

Based on the visualization, the local extrema appear to be at $x = 0$ (maximum) and $x = 2$ (minimum).

1. **Find Critical Points ($f'(x) = 0$)**

$$f(x) = x^3 - 3x^2 + 2$$
$$f'(x) = 3x^2 - 6x$$

Set the derivative to zero:

$$0 = 3x(x-2)$$
$$\implies x_1 = 0, \quad x_2 = 2$$

2. **Classify Extrema (Second Derivative Test)**

   Calculate $f''(x)$:
   $$f''(x) = 6x - 6$$

   Evaluate at critical points:

   - **At $x_1 = 0$:**
   $$f''(0) = 6(0) - 6 = -6 < 0$$
   Since $f''(0)$ is negative, $f(x)$ has a **local maximum** at $x = 0$.
   - **At $x_2 = 2$:**
   $$f''(2) = 6(2) - 6 = 6 > 0$$
   Since $f''(2)$ is positive, $f(x)$ has a **local minimum** at $x = 2$.

   Conclusion: The guess was correct. There is a local maximum at $x = 0$ and a local minimum at $x = 2$.

4 Problem 4 DONE

4.1 Task

Consider the function $f : [-2, 2] \to \mathbb{R}$ $f(x) = x^4 - 2x^2 + 1$. Determine its global extrema. Afterwards, illustrate your results in Julia.

4.2 Solution

4.2.1 Code

```
using Plots

f(x) = x^4 - 2x^2 + 1

# plotting range
x = -2.2:0.05:2.2
y = f.(x)

plot(x, y,
     title = "Global Extrema of f(x) = x^4 - 2x^2 + 1",
```

```
    xlabel = "x",
    ylabel = "y",
    label = "f(x)",
    linewidth = 2,
    legend = :top)

# highlight Global Maxima
scatter!([-2, 2], [f(-2), f(2)], label="Global Maxima", color=:red, markersize=5)

# highlight Global Minima
scatter!([-1, 1], [f(-1), f(1)], label="Global Minima", color=:blue, markersize=5)

plot!([-2,2], seriestype = :vline, label="Domain")

savefig("./figs/p_04.png")
```

4.2.2 Solution

To find the global extrema of a continuous function on a closed interval $[-2, 2]$, we must compare the function values at:

1. **Critical points** inside the interval $(-2, 2)$.
2. **Endpoints** of the interval.

1. Find Critical Points Set $f'(x) = 0$:

$$\begin{aligned} f(x) &= x^4 - 2x^2 + 1 \\ f'(x) &= 4x^3 - 4x \\ 0 &= 4x(x^2 - 1) \\ 0 &= 4x(x-1)(x+1) \end{aligned}$$

The critical points are $x \in \{0, 1, -1\}$. All lie within $[-2, 2]$.

2. Evaluate Candidates We calculate $f(x)$ for all critical points and endpoints:

Type	x	Calculation	$f(x)$
:—	:—:	:—	:—:
Endpoint	-2	$(-2)^4 - 2(-2)^2 + 1 = 16 - 8 + 1$	**9**
Critical Pt	-1	$(-1)^4 - 2(-1)^2 + 1 = 1 - 2 + 1$	**0**
Critical Pt	0	$0^4 - 2(0)^2 + 1$	1
Critical Pt	1	$1^4 - 2(1)^2 + 1$	**0**
Endpoint	2	$2^4 - 2(2)^2 + 1$	**9**

3. Conclusion Comparing the values:

- The **Global Maximum** is **9**, occurring at $x = -2$ and $x = 2$.
- The **Global Minimum** is **0**, occurring at $x = -1$ and $x = 1$.

5 Problem 5 DONE

5.1 Task

Suppose a rectangle with side length x and y has the perimeter 20. (a) Express the area A as a function of x. (b) Determine the maximal possible area.

5.2 Solution

5.2.1 (a) Express Area $A(x)$

The perimeter is given by $2x + 2y = 20$. Dividing by 2, we get the constraint:

$$x + y = 10 \implies y = 10 - x$$

The area A is the product of the sides (xy). Substituting y:

$$A(x) = x(10 - x) = 10x - x^2$$

Since side lengths must be non-negative ($x \geq 0$ and $y = 10 - x \geq 0$), the domain is $x \in [0, 10]$.

$$A : [0, 10] \to \mathbb{R}, \quad A(x) = -x^2 + 10x$$

5.2.2 (b) Determine Maximal Area

To find the maximum, we look for critical points where the derivative is zero.

$$\begin{aligned} A(x) &= -x^2 + 10x \\ A'(x) &= -2x + 10 \\ 0 &= -2x + 10 \\ 2x &= 10 \\ x &= 5 \end{aligned}$$

We verify this against the endpoints of the domain $[0, 10]$:

- $A(0) = 0$ (Minimum)
- $A(10) = 0$ (Minimum)
- $A(5) = 5(10 - 5) = 25$ (Maximum)

Conclusion: The maximal possible area is **25** (occurring when the rectangle is a square with $x = 5$).